REVIEW · NUTRITION SCIENCE

Are seed oils really bad for you?

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ABSTRACT

For generally healthy adults, fresh, unhydrogenated seed oils are reasonable culinary fats when they replace foods rich in saturated fat. Controlled feeding trials consistently lower low-density lipoprotein (LDL) cholesterol and generally improve short-term glucose–insulin markers; prospective intake and biomarker studies are mostly neutral or favorable, and randomized trials do not show a consistent rise in systemic inflammatory markers. One synthesis estimated a 9.83 mg/dL LDL reduction (95% confidence interval [CI]: 6.04–13.63 mg/dL lower) with a higher ratio of polyunsaturated to saturated fat. Clinical cardiovascular-event trials remain old and discordant, so they cannot establish a precise event benefit, but they do not support a large class-wide hazard. This answer is conditional on the replacement food and preparation: adding oil as surplus energy is different from substitution, repeatedly heated frying oil accumulates oxidation products, chronic human risk from those products is not quantified, and an oil ingredient cannot explain the risk of an entire ultra-processed food pattern.

01 Introduction

“Seed oils” is a culinary and internet category, not a standardized exposure.[1] It usually includes soybean, corn, sunflower, safflower, canola, grapeseed, rice-bran, and cottonseed oils, but those oils differ in linoleic acid, alpha-linolenic acid, oleic acid, saturated fat, antioxidants, cultivar, and refining.[2,1] Some newer cultivars are high-oleic—bred to contain more oleic acid.[1,3] Some foods contain fresh oil; others contain oil that has spent hours in a commercial fryer.[1] A bottle used to sauté vegetables once is biologically different from repeatedly reused deep-frying oil, and both are different from a packaged food whose risk profile also includes energy density, starch, salt, sugar, additives, texture, and eating pattern.[1,3]

This review therefore treats the claim “seed oils are bad” as a set of testable questions. What happens when unsaturated oil replaces butter or another saturated-fat-rich food? Do linoleic-acid interventions raise inflammatory markers? Do intake and tissue biomarkers predict cardiovascular disease, diabetes, cancer, or death? What is established about heating and oxidation? The distinction between replacing and adding fat is central: an isocaloric (equal-calorie) substitution can improve a risk marker even though adding the same oil on top of an energy-surplus diet could promote weight gain.[4,5] The food matrix and processing context can also modify outcomes independently of the oil.[6,5]

The hierarchy of evidence differs by outcome. Randomized feeding trials provide direct evidence for lipids and short-term metabolic markers.[7,8] Clinical-event trials are harder to interpret because they were performed decades ago with mixed interventions, institutional diets, imperfect adherence, and sometimes trans-fat-containing margarines.[9,10,11] Large prospective cohorts and biomarker consortia add duration and event numbers but cannot eliminate confounding.[12] Mechanistic and thermal-chemistry experiments establish plausibility, not the size of chronic human risk.[13,14] The conditional answer therefore depends on matching each claim to the design that can test it.

02 “Seed oils” does not name one biological exposure

The first failure of the blanket claim is compositional. Reviews of edible oils distinguish linoleic-rich oils from high-oleic oils, oils that contain appreciable alpha-linolenic acid, and relatively saturated plant fats.[2,1] Refining removes some minor compounds and changes flavor and stability; breeding can shift the fatty-acid profile enough that two sunflower oils behave differently under heat.[1,3] The umbrella evidence accordingly ranges from moderate to very low certainty across oil–outcome pairs rather than licensing a class-wide verdict.[1,3,15]

Hydrogenation is another essential boundary. Older partially hydrogenated shortenings and margarines could contain trans fatty acids, whose molecular geometry and lipid effects differ from the ordinary non-trans form of linoleic acid.[16,4] Trials comparing linoleic acid, stearic acid, and trans fat demonstrate why historical products cannot simply be relabeled as modern unhydrogenated oil.[16,4] “Omega-6” is also broader than linoleic acid: its metabolites need not share the same associations, and tissue levels reflect endogenous conversion—metabolism within the body—as well as diet.[17,18]

The evidence is most interpretable when exposure is specified along three axes: the named oil or fatty acid, what it replaces, and how the oil-containing food is prepared. Controlled feeding establishes short-term marker responses; cohorts establish associations; heating experiments establish chemistry; none can silently substitute for the others.[19,12,13,20]

03 The replacement food determines the direction of effect

Dietary fats do not enter a vacuum. If oil replaces butter, saturated fat falls and polyunsaturated or monounsaturated fat rises.[21,22] If oil replaces another unsaturated oil, the contrast can be small.[22,23] If oil is added without reducing anything else, total energy rises.[23] These are different interventions, and the literature changes direction accordingly.[21,22,23]

Classic metabolic-ward and controlled-feeding work established that exchanging saturated for unsaturated fat improves total and LDL cholesterol more reliably than merely reducing fat.[24,25] Whole-food comparisons reached the same broad conclusion while also revealing matrix effects: soybean oil, olive oil, dairy butter, cocoa butter, walnuts, and matched vegetable-oil diets do not collapse perfectly onto their fatty-acid totals.[26,5] In the Dietary Intervention and Vascular Function (DIVAS) trial, replacing saturated with unsaturated fat improved lipid biomarkers and blood pressure without a detectable vascular-function change, an example of a favorable marker result that should not be inflated into a clinical-event claim.[27]

The practical implication is narrow but important. A spoonful of unhydrogenated soybean, corn, canola, sunflower, or safflower oil used instead of butter is supported by a different evidence base from the same spoonful added to a deep-fried, energy-dense meal.[4,5] “What did it replace?” is not a statistical technicality; it identifies the intervention.

04 Replacing saturated fat with unsaturated oil lowers LDL cholesterol

The most consistent human result concerns LDL cholesterol.[28,7] A network meta-analysis, which combines direct and indirect treatment comparisons, of 54 randomized trials found that most unsaturated oils lowered LDL by roughly 0.23–0.42 mmol/L compared with butter.[28] A more recent synthesis of 24 controlled-feeding trials estimated that a higher ratio of polyunsaturated to saturated fat lowered LDL by 9.83 mg/dL, with a 95% CI from −13.63 to −6.04 mg/dL.[7] Trials of corn oil, walnuts or matched oils, and saturated-to-unsaturated replacement agree on direction even when their secondary outcomes differ.[29,5]

Direct linoleic-acid trials show a smaller effect because their comparators often contain other unsaturated fats.[30,15] Across 40 randomized trials and 2,175 participants, higher linoleic acid reduced LDL by 3.26 mg/dL (95% CI −5.78 to −0.74) and slightly reduced high-density lipoprotein cholesterol, while total cholesterol and triglycerides were not clearly changed.[30] Reviews of healthy adults and high-oleic substitutions make the same comparator-dependent point.[15]

Figure 1 compares the two estimates on their reported mg/dL scale.

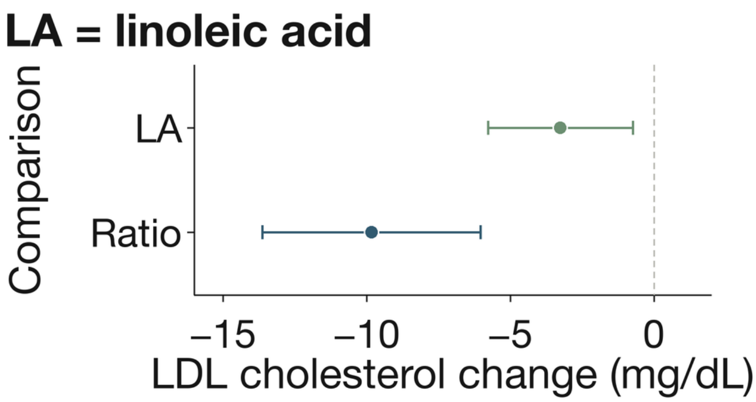

Figure 1. Unsaturated-fat substitutions lower LDL cholesterol. Top: direct linoleic-acid estimate, –3.26 mg/dL (95% CI –5.78 to –0.74). Bottom: higher-versus-lower polyunsaturated-to-saturated-fat ratio, –9.83 mg/dL (–13.63 to –6.04). The x-axis shows low-density lipoprotein cholesterol change; whiskers are 95% confidence intervals and the dashed line is no difference. Interventions and comparators differ.[7,30,28]

These are surrogate outcomes, usually measured over weeks.[7,30] Low-density lipoprotein cholesterol is causally relevant to atherosclerotic risk, but a short trial cannot itself show that an oil prevents myocardial infarction.[4,7] That distinction becomes decisive in the older clinical-event literature.

05 Clinical-event trials disagree about the size of cardiovascular benefit

One meta-analysis of eight randomized trials, 13,614 participants, and 1,042 coronary events estimated a coronary-event relative risk (RR) of 0.81 (95% CI 0.70–0.95) when polyunsaturated fat replaced saturated fat.[31] Expressed per 5% of energy replaced, the estimate was 0.90.[31] Cochrane syntheses similarly conclude that reducing saturated fat or increasing polyunsaturated fat probably reduces cardiovascular events, although effects on total mortality are small or uncertain.[32,33,34]

The estimate changes when analysts restrict the trial set.[35] A review limited to trials judged adequately controlled estimated major coronary events at 1.06 (0.86–1.31), compatible with benefit or harm.[35] An earlier analysis argued that interventions specific to omega-6 linoleic acid should be separated from mixtures that also raised omega-3 fats.[36] The disagreement reflects trial selection, intervention composition, adherence, and outcome definition—not merely random noise.[35,36]

Recovered data sharpened the uncertainty. In the Sydney Diet Heart Study, the linoleic-acid intervention group had mortality of 17.6% versus 11.8%, hazard ratio (HR) 1.62 (1.00–2.64).[9] In the Minnesota Coronary Experiment, serum cholesterol fell by 13.8% without a mortality or autopsy benefit; the accompanying meta-analysis estimated coronary-heart-disease mortality at 1.13 (0.83–1.54).[10,11] Both trials involved historically specific institutional or secondary-prevention settings.[9,10,11] Other legacy trials—the Los Angeles Veterans trial and Oslo Diet-Heart Study among them—favored coronary outcomes, while a small corn-oil trial did not.[37,38,39]

Figure 2 places four headline estimates on one axis while preserving their non-equivalence.

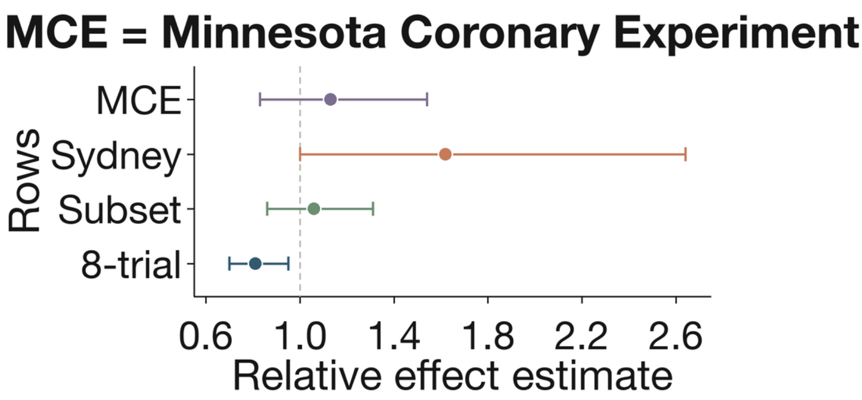

Figure 2. Legacy clinical-event estimates disagree. Top to bottom: Minnesota-linked coronary mortality, 1.13 (95% CI 0.83–1.54); Sydney all-cause mortality, 1.62 (1.00–2.64); adequately controlled subset, 1.06 (0.86–1.31); favorable eight-trial coronary-event pool, 0.81 (0.70–0.95). The x-axis gives the reported relative effect; whiskers are 95% confidence intervals and 1.0 is the null. Outcomes and designs differ.[31,35,9,10]

The trials exclude a many-fold general benefit or hazard more confidently than they identify a small effect.[31,35,9,10] Their original reports span 1965–1989 and tested historically specific diets or products, so they predate statins, contemporary hypertension care, and current oil formulations.[39,37,38,11] The Women's Health Initiative low-fat pattern adds scale but does not isolate seed oil or an isocaloric saturated-fat substitution.[40] A modern long-duration trial of named unhydrogenated oils would be more informative than another reanalysis of mixed mid-century interventions.

ANALYSIS	ESTIMATE	BOUNDARY
Pooled[31]	Relative risk 0.81	Mixed trials
Strict[35]	Relative risk 1.06	Low power
Sydney[9]	Hazard ratio 1.62	Historical product
Minnesota[10]	Relative risk 1.13	Institutional exposure

06 Prospective intake and biomarker evidence does not support broad vascular harm

Large prospective cohorts generally associate linoleic acid, plant fat, or plant oils with neutral or lower cardiovascular and mortality risk.[41,42,12] A meta-analysis of dietary linoleic acid, the Nurses' Health Study, and mortality cohorts point in a similar direction when polyunsaturated fat replaces saturated fat.[41,43,42] A 521,120-person cohort likewise found different mortality associations for specific dietary fats rather than a single class effect.[44]

Recent oil-level analyses are unusually direct, though still observational. Across three US cohorts with repeated diet measures, the highest plant-oil intake was associated with total-mortality HR 0.84 (0.79–0.90), and replacing 10 g/day of butter with plant oil was associated with HR 0.83 (0.79–0.86).[45] In 407,531 Diet and Health Study participants, vegetable-oil fat was associated with HR 0.88 for total and 0.85 for cardiovascular mortality, and modeled replacement of animal fat with plant fat was favorable.[46] Healthy-user confounding and food-frequency error remain plausible, so these estimates should not be sold as prescription effects.[45,46]

Biomarkers reduce some self-report error.[12] A harmonized consortium of 30 prospective studies, 68,659 participants, and 15,198 events associated higher circulating or tissue linoleic acid with total cardiovascular HR 0.93 (0.88–0.99), cardiovascular-mortality HR 0.78 (0.70–0.85), and ischemic-stroke HR 0.88 (0.79–0.98).[12] Other biomarker cohorts are consistent.[47] In Danish adipose tissue, the highest versus lowest linoleic-acid quartile predicted total ischemic-stroke HR 0.78 (0.65–0.93) and large-artery stroke HR 0.61 (0.43–0.88).[48]

Figure 3 summarizes representative prospective estimates. It should be read as consistency across observational lenses, not as proof that adding oil prevents disease.

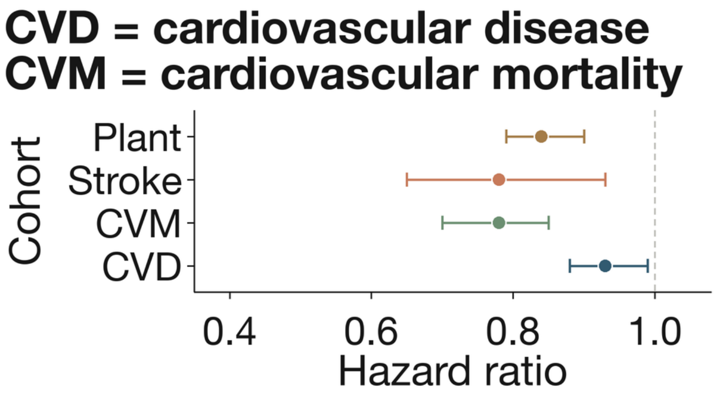

Figure 3. Prospective estimates point below the null. Top to bottom: plant-oil intake and mortality, hazard ratio 0.84 (95% CI 0.79–0.90); adipose linoleic acid and ischemic stroke, 0.78 (0.65–0.93); circulating or tissue linoleic acid and cardiovascular mortality, 0.78 (0.70–0.85); and total cardiovascular disease, 0.93 (0.88–0.99). The x-axis is the hazard ratio; whiskers are 95% confidence intervals and 1.0 is the null. All are observational.[12,48,45]

A Mendelian-randomization study, which uses genetic variants as proxies for an exposure, adds a different limitation. Genetically predicted linoleic acid was not associated with ischemic heart disease across up to 76,014 cases and 264,785 controls, even though the instruments predicted slightly lower diabetes odds and cholesterol.[49] Lifelong fatty-acid metabolism is not the same intervention as replacing butter with oil, but the null result argues against an easily detectable direct effect of large magnitude.[49]

07 Randomized trials do not show a consistent rise in systemic inflammation

The common mechanistic story is that dietary linoleic acid supplies arachidonic-acid pathways and therefore drives chronic inflammation.[50,51] Human intervention data do not support that simple chain.[50,52] A systematic review of 15 trials found no consistent increase in C-reactive protein, cytokines, fibrinogen, adhesion molecules, or tumor necrosis factor.[50] A later meta-analysis of 30 randomized trials and 1,377 participants estimated C-reactive protein standardized mean difference (SMD) 0.09 (−0.05 to 0.24), interleukin-6 0.11 (−0.07 to 0.29), and tumor-necrosis-factor SMD −0.01 (−0.19 to 0.17).[52]

Figure 4 shows why "no consistent increase" is more accurate than "anti-inflammatory".[52,50] Every interval crosses zero, and the trials were generally short and often enrolled healthy adults.[52,50]

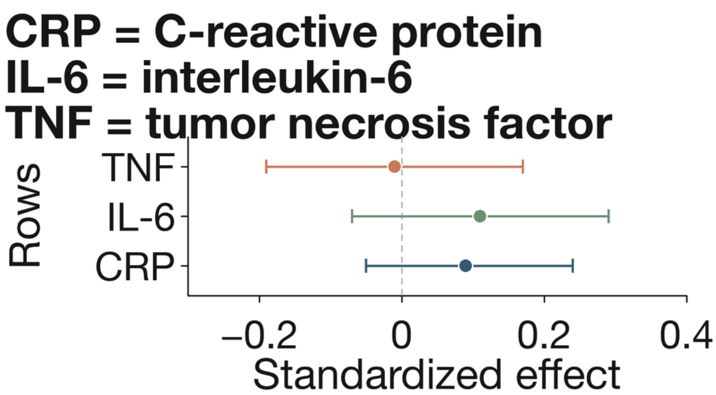

Figure 4. Randomized inflammation-marker estimates cross the null. Top to bottom: tumor necrosis factor, standardized mean difference −0.01 (95% CI −0.19 to 0.17); interleukin-6, 0.11 (−0.07 to 0.29); C-reactive protein, 0.09 (−0.05 to 0.24). The x-axis gives standardized effects; whiskers are 95% confidence intervals and zero is no difference. Short trials do not exclude tissue-specific or long-latency effects.[52,50,51]

Human feeding data also weaken the assumed precursor chain. Across trials that decreased linoleic acid by as much as 90% or increased it up to six-fold, changes in dietary linoleic acid did not correlate with plasma, serum, or erythrocyte arachidonic acid.[53] Observational studies of plasma fatty acids and habitual intake report heterogeneous inflammatory associations rather than a uniform omega-6 signal.[54,55]

Mechanistic caution is still warranted. Reviews describe oxidized linoleic-acid metabolites and context-dependent inflammatory pathways, and advocates of the oxidized-linoleic hypothesis identify plausible vascular mechanisms.[51,56] Plausibility is strongest where oxidation is demonstrated—especially prolonged high heat—not as a reason to override randomized systemic-marker results for ordinary fresh-oil intake.[51,13]

08 Metabolic evidence is favorable for markers but not definitive for disease prevention

In a meta-analysis of 102 controlled feeding trials, 239 diet arms, and 4,220 adults, replacing 5% of energy from carbohydrate with polyunsaturated fat lowered glycated hemoglobin, a marker of average blood glucose, by 0.11 percentage points (95% CI −0.17 to −0.05) and lowered fasting insulin.[8] Replacing saturated fat with polyunsaturated fat improved glucose, glycated hemoglobin, C-peptide, and homeostatic measures.[8] These are short-term markers, not diabetes incidence.[8]

Longer-duration randomized evidence is less decisive.[57] A synthesis of 83 randomized trials found little or no effect of omega-6 or total polyunsaturated fat on glycemic markers and very uncertain evidence for new diabetes; only 10 trials were at low summary risk of bias.[57] The apparent tension is explained partly by different interventions: tightly controlled equal-calorie feeding can detect small physiological changes that heterogeneous long trials were not designed to isolate.[8,57]

Prospective biomarkers are consistently favorable.[17,58] In the European Prospective Investigation into Cancer and Nutrition–InterAct (EPIC-InterAct) study, plasma linoleic acid predicted incident-diabetes HR 0.80 (0.77–0.83) per standard deviation.[17] A pooled analysis of 20 cohorts and 39,740 adults estimated RR 0.65 (0.60–0.72) across the interquintile biomarker range.[58] Intake meta-analyses likewise associate linoleic acid or vegetable fat with lower diabetes risk, with the ordinary caveats of diet measurement and residual confounding.[59,60]

Figure 5 displays the two biomarker estimates. Their precision should not conceal their observational design.[17,58]

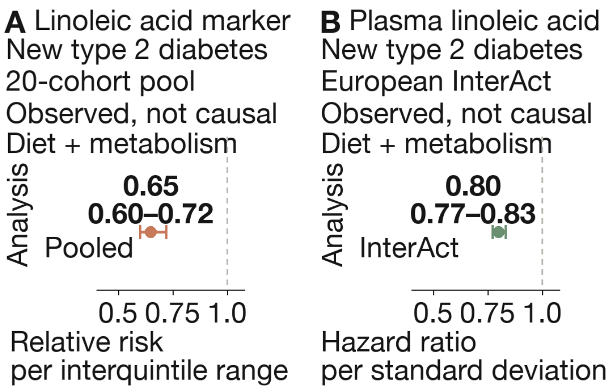

Figure 5. Linoleic-acid biomarkers are associated with lower diabetes incidence. Top: 20-cohort pool, relative risk 0.65 (95% CI 0.60–0.72) per interquintile range. Bottom: European InterAct cohort, hazard ratio 0.80 (0.77–0.83) per standard deviation of plasma linoleic acid. The x-axis gives the reported relative effect; whiskers are 95% confidence intervals and 1.0 is the null. These are observational, noncausal estimates on different scales, and biomarkers reflect intake plus metabolism.[17,58,59]

Controlled overfeeding provides a complementary physiological signal. In the Metabolic Consequences of Moderate Weight Gain–Role of Dietary Fat Composition (LIPOGAIN) trial, equal weight gain produced about 50% more liver fat with saturated fat but not sunflower-oil polyunsaturated fat; a later trial replicated the divergence in liver fat and ceramides, a class of lipid molecules.[61,62] In adults with prediabetes or type 2 diabetes, a 12-month polyunsaturated-fat-rich diet reduced liver fat by 1.46 percentage points versus usual care, although a healthy Nordic diet improved a wider set of outcomes.[63] These trials argue that fat type matters under controlled energy surplus; they do not make overfeeding harmless.[61,62,63]

09 Cancer evidence excludes a large general effect but leaves site-specific uncertainty

Cancer is not one outcome, and most fat trials were not designed or powered for it.[64] A meta-analysis of 47 randomized trials and 108,194 participants found little or no overall cancer effect for omega-3 and a small, imprecise signal for total polyunsaturated fat: RR 1.19 (0.99–1.42), corresponding to a reported number needed to harm—the number of people who would need the exposure for one additional outcome—of 125 if causal.[64] Prospective syntheses of dietary fat and mortality are mostly neutral or favorable overall.[65,66]

Site-specific observational results remain mixed.[67,68,66] A colorectal-cancer meta-analysis of 20 cohorts and 787,490 participants estimated dietary linoleic-acid RR 1.15 (1.05–1.27) for highest versus lowest intake, while tissue linoleic acid was null at RR 0.93 (0.61–1.41).[67] A prostate-cancer biomarker synthesis did not yield a class-wide answer, and other analyses report possible ovarian or endometrial signals that need replication.[68,66]

The discordance between reported intake and tissue biomarkers is an evidence warning.[67] It could reflect measurement error, food context, metabolism, confounding, or genuine differences by cancer site.[67,68] It does not justify claiming that ordinary seed-oil use generally causes cancer, nor does it prove absence of every site-specific effect.

The conditional answer and its main preparation boundary are summarized in Figure 6.

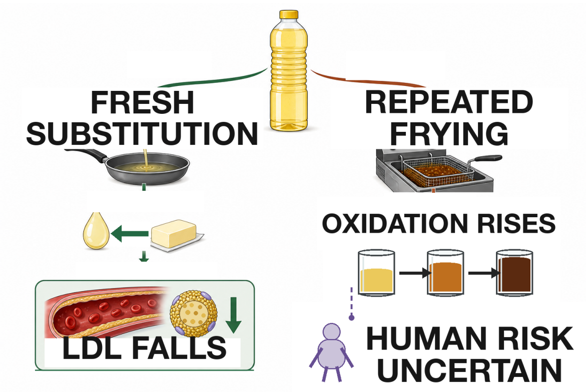

Figure 6. Fresh substitution and repeated frying are different exposures. Fresh unsaturated oil replacing saturated fat lowers low-density lipoprotein cholesterol. Repeated high-temperature reuse raises oxidation products, but long-term human clinical risk remains unquantified.[7,14,69]

10 Repeated high-temperature frying is a distinct and credible hazard boundary

Thermal chemistry is where the strongest seed-oil-specific concern resides.[13,69] Heating lipids from 100 to 200°C increases multiple oxidation products, and the pattern depends on fatty-acid composition.[13] In one controlled comparison, soybean oil at 200°C had 3.15 micrograms per gram of malondialdehyde, a lipid-oxidation product, and substantially higher signals for the volatile aldehydes hexanal and 2,4-heptadienal than comparator oils.[13] Reviews of frying describe thermoxidation (heat-driven oxidation), polymerization (molecules joining into larger products), hydrolysis (water-driven bond cleavage), antioxidant depletion, and, under severe conditions, isomerization (rearrangement into a different molecular form); high-oleic composition, food moisture, oxygen, time, and replenishment all change the rate.[69,70,71]

Repeated frying studies show that aldehydes can accumulate in oil and transfer into food.[72,14] Restaurant fries have contained roughly 10–25 parts per million per monitored aldehyde class in one experimental program, while animal feeding studies report tissue injury after repeatedly heated oil.[72,14,73] Oxidized linoleic-acid metabolites can activate a pain receptor in rodents, establishing biological activity but not a human chronic-disease dose.[74]

Human outcome evidence is sparse.[14,69] A 16-person crossover study found that meals rich in heated oils acutely shortened the lag time to serum oxidation by about 12%, but it did not measure disease.[75] Prospective human studies have not yet quantified clinical risk against individual exposure histories combining time–temperature, fryer reuse, and absorbed oxidation-product dose.[14,69] The rational boundary is therefore exposure reduction without false precision: avoid repeatedly reusing deep-frying oil; do not infer a universal “safe temperature” from smoke point alone; and do not treat fresh oil used once as equivalent to an industrial or restaurant fryer cycle.

CONTEXT	FINDING	BOUNDARY
200°C[13]	Oxidation rises	Chemistry
Repeated frying[71]	Oil degrades	Limit reuse
Fried food[72]	Aldehydes transfer	Dose unknown
Human meal[75]	Acute oxidation	No disease outcome

11 Ultra-processed-food risk cannot be assigned to the oil ingredient

An umbrella review associated higher ultra-processed-food exposure with cardiovascular, metabolic, and mortality outcomes, but certainty was often low or very low and the exposure bundled many food properties.[20] Classification critiques likewise emphasize confounding by overall diet and difficulty separating formulation from processing.[76,77,78] In UK Biobank, cardiovascular associations differed between plant-origin ultra-processed and non-ultra-processed foods;[79] in the European Prospective Investigation into Cancer and Nutrition, diabetes associations varied across ultra-processed subgroups.[80] These designs do not isolate an oil ingredient, so an oil-containing packaged food and unsaturated oil replacing butter are not equivalent exposures.[79,80]

12 The omega-6:omega-3 ratio can mislead even when it predicts risk

Ratios compress two variables into one: a higher omega-6:omega-3 ratio could mean higher omega-6, lower omega-3, or both. In UK Biobank, the highest versus lowest plasma ratio predicted all-cause mortality HR 1.26 (1.15–1.38), cancer mortality 1.14 (1.00–1.31), and cardiovascular mortality 1.31 (1.10–1.55).[81] Yet both component concentrations were separately inversely associated with all-cause mortality.[81]

Figure 7 shows the ratio association, not a causal effect of omega-6.

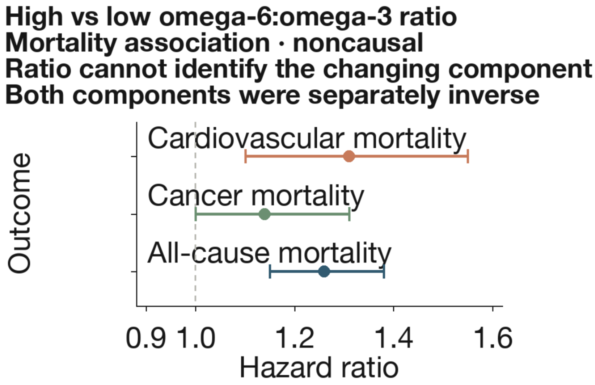

Figure 7. A high omega-6:omega-3 ratio is associated with mortality but not its cause. Top to bottom: cardiovascular mortality, hazard ratio 1.31 (95% CI 1.10–1.55); cancer mortality, 1.14 (1.00–1.31); all-cause mortality, 1.26 (1.15–1.38). The x-axis is the hazard ratio; whiskers are 95% confidence intervals and 1.0 is the null. This is an observational, noncausal comparison. Both component concentrations were separately inverse, so the ratio cannot identify which component changed and lowering omega-6 is not the sole interpretation.[81,12,17]

Earlier cohort work and reviews proposed ratio targets or linked higher ratios with mortality.[82,83,18] A ratio may be descriptive, but an intervention must specify which component changes.

13 Conclusion

The strongest version of the claim—common unhydrogenated seed oils are intrinsically toxic and should be avoided as a class—is not supported by the human evidence.[1,3] When unsaturated oils replace saturated-fat-rich foods, LDL cholesterol falls consistently.[28,7] Short-term glucose–insulin markers generally improve, prospective intake and linoleic-acid biomarkers are neutral to favorable, and randomized trials do not show a consistent rise in systemic inflammatory markers.[8,12,52] Clinical cardiovascular-event trials are too old and discordant to specify a precise benefit, but they do not establish a large general hazard.[31,35,9,10]

The boundaries matter. Do not add oil as surplus energy,[4,5] equate partially hydrogenated historical products with modern liquid oils,[16,4] or infer an isolated oil effect from whole ultra-processed-food patterns.[20,76] Minimize repeatedly reused high-temperature frying oil, although chronic human risk remains unquantified.[14,69] Cancer signals remain site-specific and observational,[67,68] and omega-6:omega-3 ratios should not replace examination of both components.[81,18]

For generally healthy adults, use fresh unsaturated oil in place of saturated-fat-rich fats when appropriate, vary fat sources, avoid repeatedly reused frying oil, and judge the whole food. Modern trials of named oils with clinical events, plus prospective measurement of frying history and oxidation-product exposure, would resolve the largest uncertainties.

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83 · VERIFIED VIA CROSSREF

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